CS2201 Computer Organization & Architecture

L23: Design Concepts in Cache Memory



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Block Replacement Algorithms

Consider a 4-way associative cache that can be operated in one among the two modes at a time. In mode-1 it uses pseudo LRU block replacement policy and in mode-2 it uses Last In First Out block replacement policy. Assume all the cache blocks are initially empty and filling up of empty blocks in a given cache set happens from way 0 to way-3. Consider the following 14 block numbers all mapped to a particular set n given in the order of arrival. A, B, C, D, A, B, E, F, A, B, F, C, D, A.

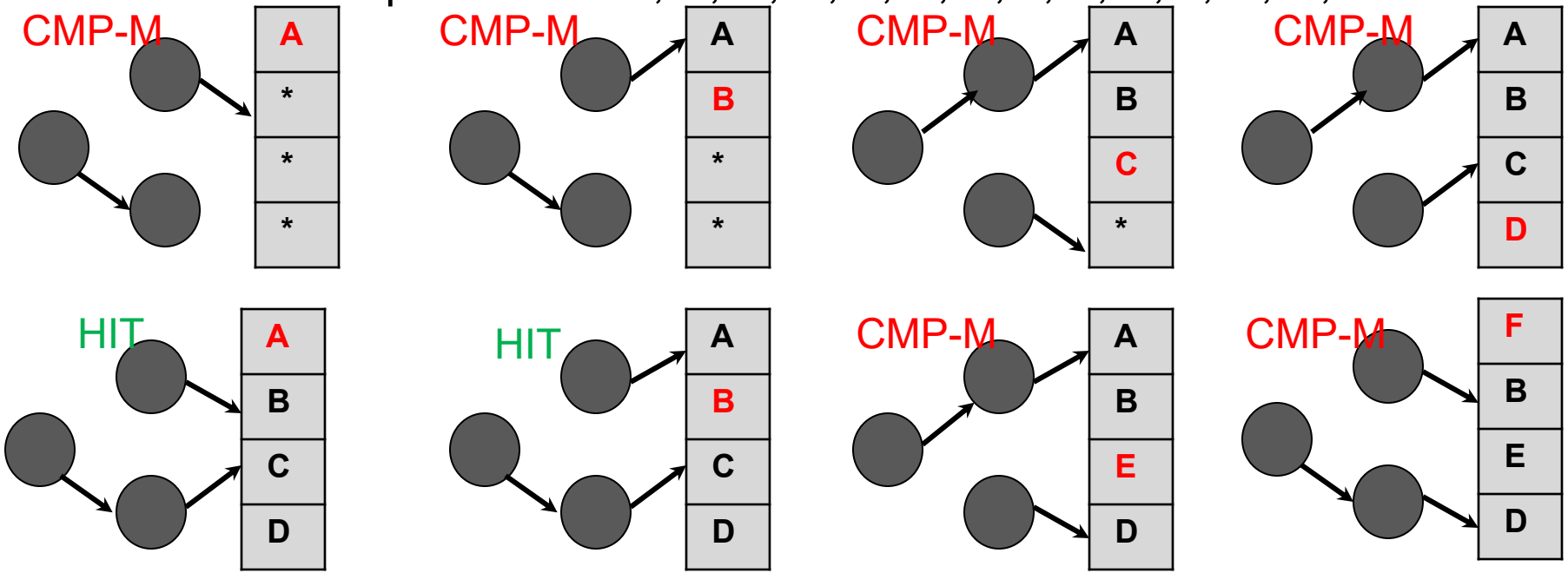
a. Find the number of cache misses in mode-1. Draw the pseudo LRU tree for set n after processing these requests in mode-1.
b. Find the number of cache misses in mode-2. Draw the content of set n (way-0 to way-3) after processing these requests in mode-2.

Block Replacement Algorithms

All the cache blocks are initially empty and filling up of empty blocks in a given cache set happens from way 0 to way-3.

pseudo LRU block replacement policy

Block List that maps to set n. A, B, C, D, A, B, E, F, A, B, F, C, D, A.

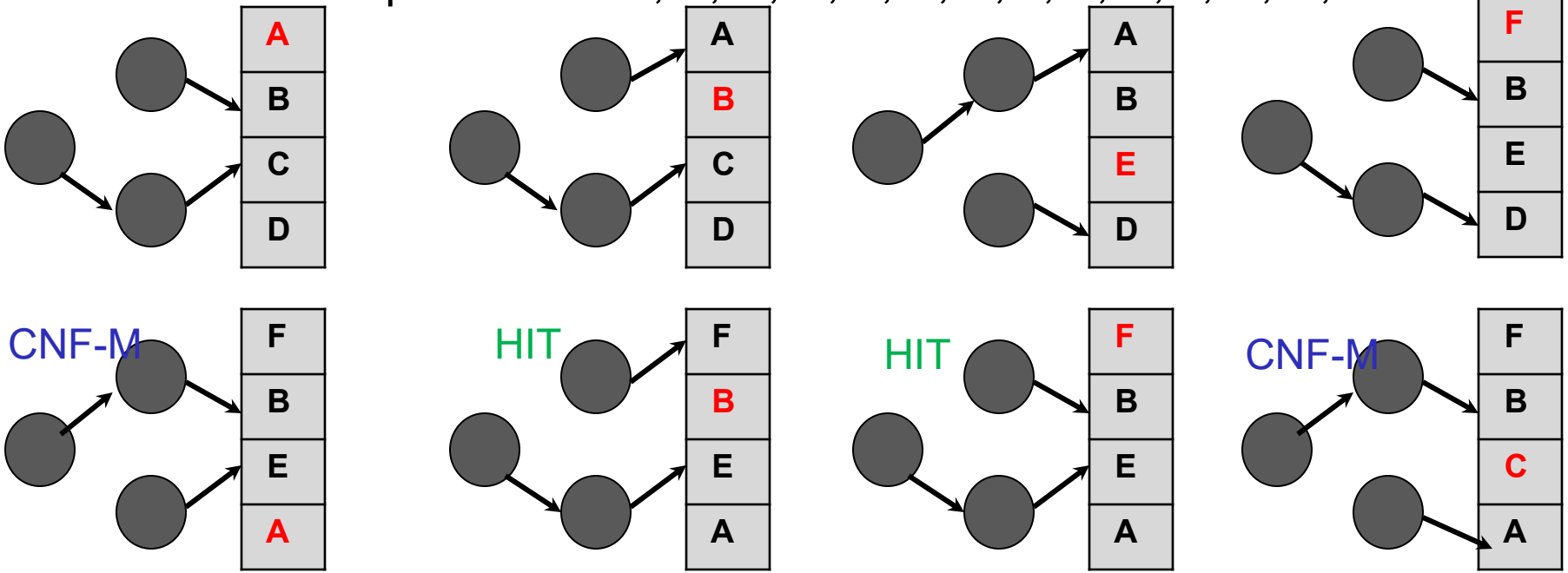


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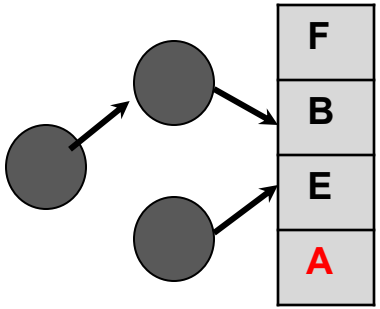


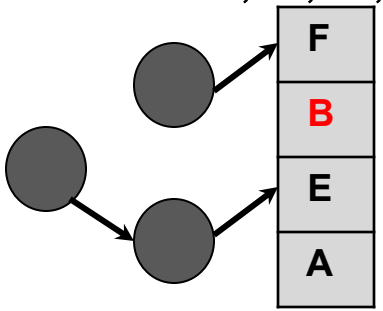
Block Replacement Algorithms

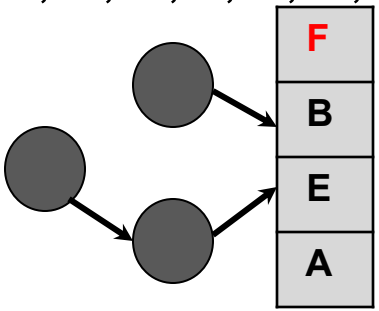
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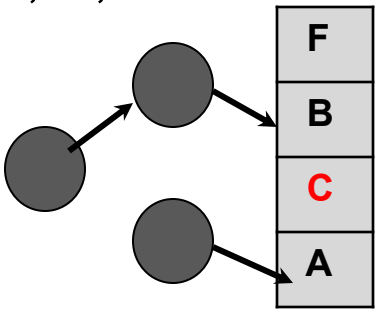
pseudo LRU block replacement policy

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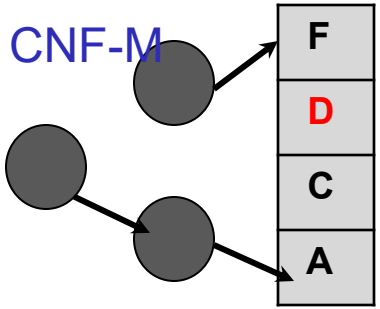




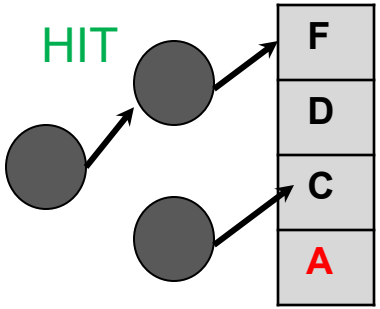




CNF-M



HIT



Total 14 access

6 CMP-Miss

3 CNF-Miss

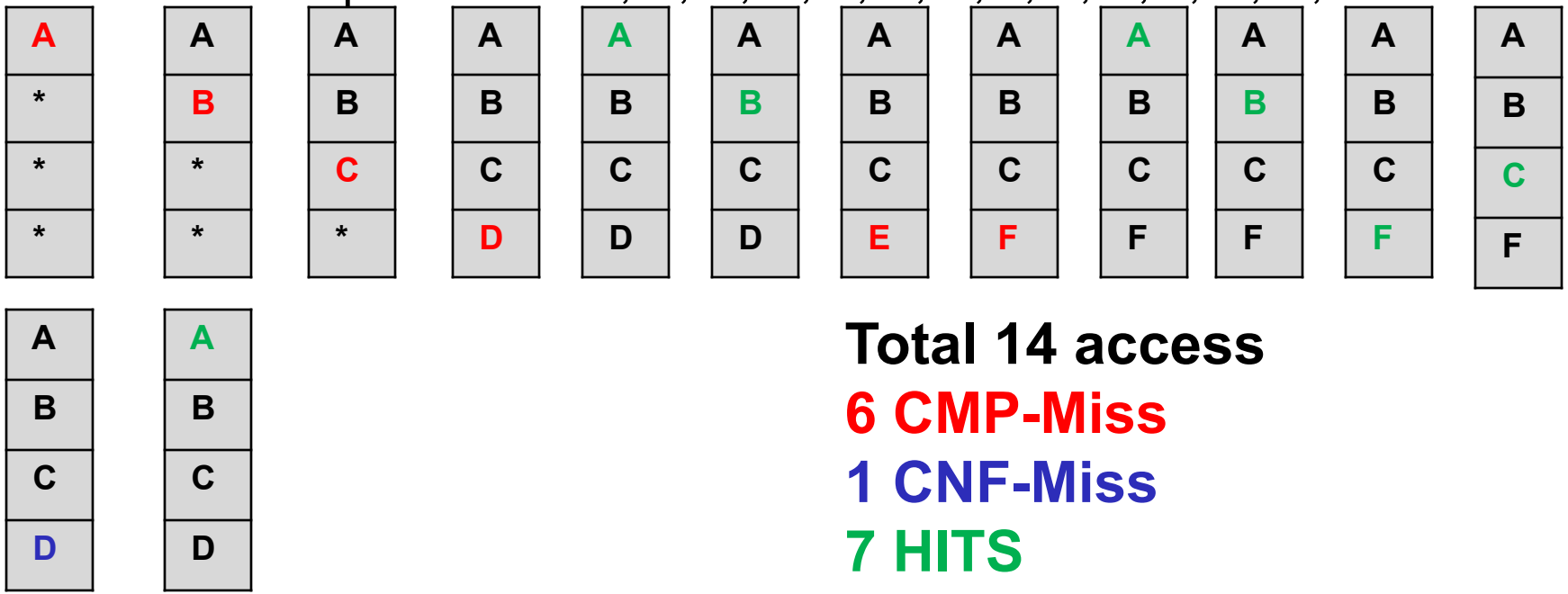
5 HITS

Block Replacement Algorithms

All the cache blocks are initially empty and filling up of empty blocks in a given cache set happens from way 0 to way-3.

LIFO block replacement policy

Block List that maps to set n. A, B, C, D, A, B, E, F, A, B, F, C, D, A.



Total 14 access

6 CMP-Miss

1 CNF-Miss

7 HITS

Cache Mapping

Consider a computer system based on 32 bit processor with 8KB direct mapped on-chip I-cache and 16KB 2-way set associative on-chip D-cache. The off-chip unified cache is 128 KB 4-way set associative. Block size for on-chip cache is 8 words and off-chip cache is 16 words. Four fixed length instruction starting at main memory word address 22 (in decimal) are executed. These instructions refer to data at main memory word address 260, 261 and 275 (all in decimal). Assuming caches are initially empty; indicate the non empty blocks on all the caches after execution of the instruction.

Instruction Words 22,23,24,25
Data 260,261,275

Cache Mapping

Instruction Words 22,23,24,25
Data 260,261,275
Location in MM @ L2 block size →**22/16, 23/16, 24/16, 25/16=B1**
Location in MM @ L2 block size → **260/16, 261/16=B16, 275/16=B17**

sets L2 = CS/(BSxA) = $2^{17}/(2^4.2^2x2^2) = 2^9$ = 512 sets

B1%512 → S1
B16%512 → S16
B17%512 → S17

One block of sets S1, S16, S17 WILL BE NON EMPTY

Cache Mapping

Instruction Words 22,23,24,25; Data 260,261,275
Location in MM @ L1-I block size → 22/8, 23/8 = B2
Location in MM @ L1-I block size → 23/8, 24/8 = B3
Location in MM @ L1-D block size → 260/8, 261/16=B32, 275/8=B34

sets L1-I = CS/(BSxA) = $2^{13}/(2^3.2^2x1) = 2^8$ = 256 sets
B2%256 → S2
B3%256 → S3
Sets S2 & S3 of LI-I cache WILL BE NON EMPTY

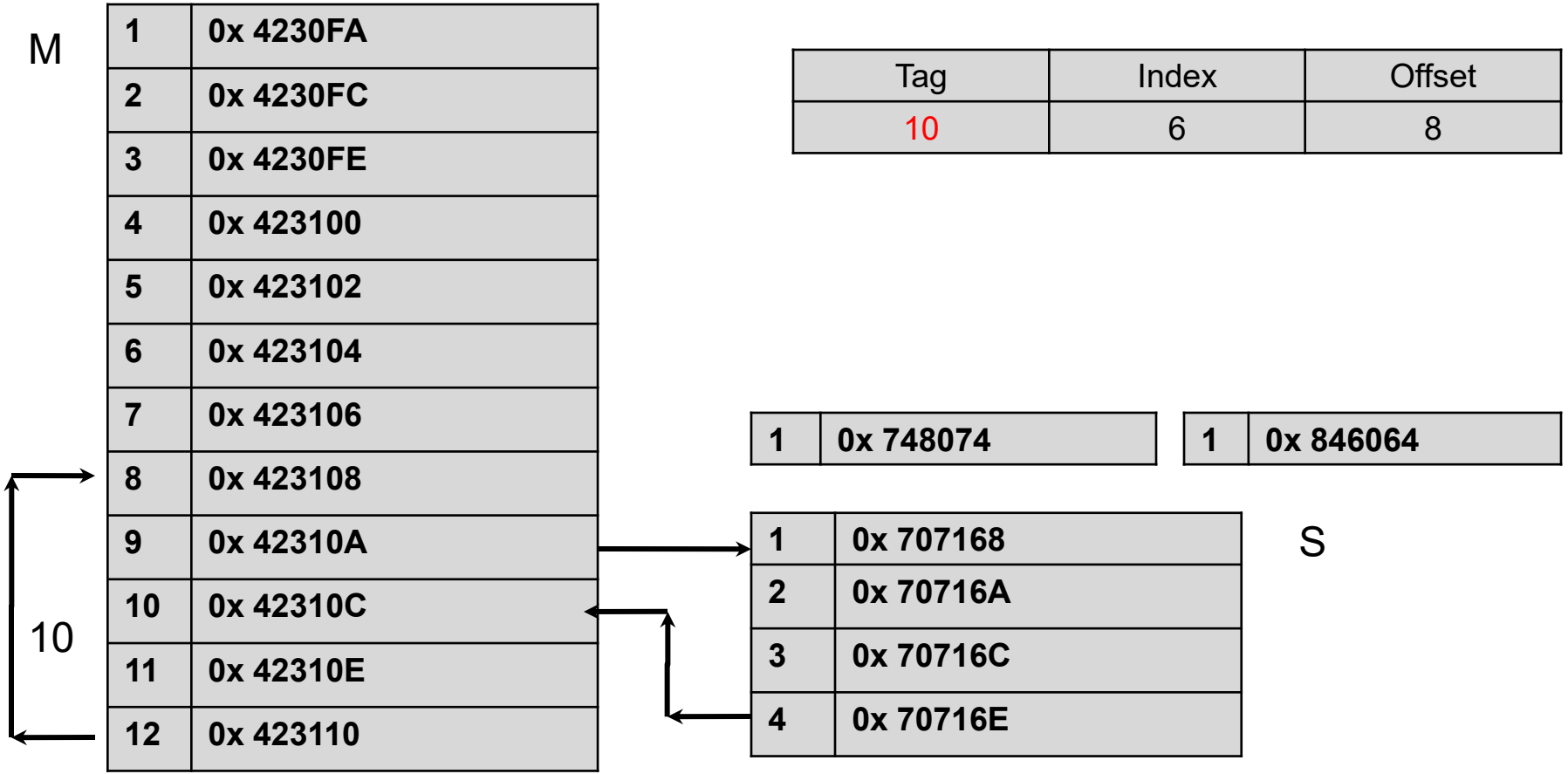
sets L1-D = CS/(BSxA) = $2^{14}/(2^3.2^2x2^1) = 2^8$ = 256 sets
B32%256 → S32
B34%256 → S34
One block of Sets S32,S34 of LI-D cache WILL BE NON EMPTY

Cache Mapping by OS

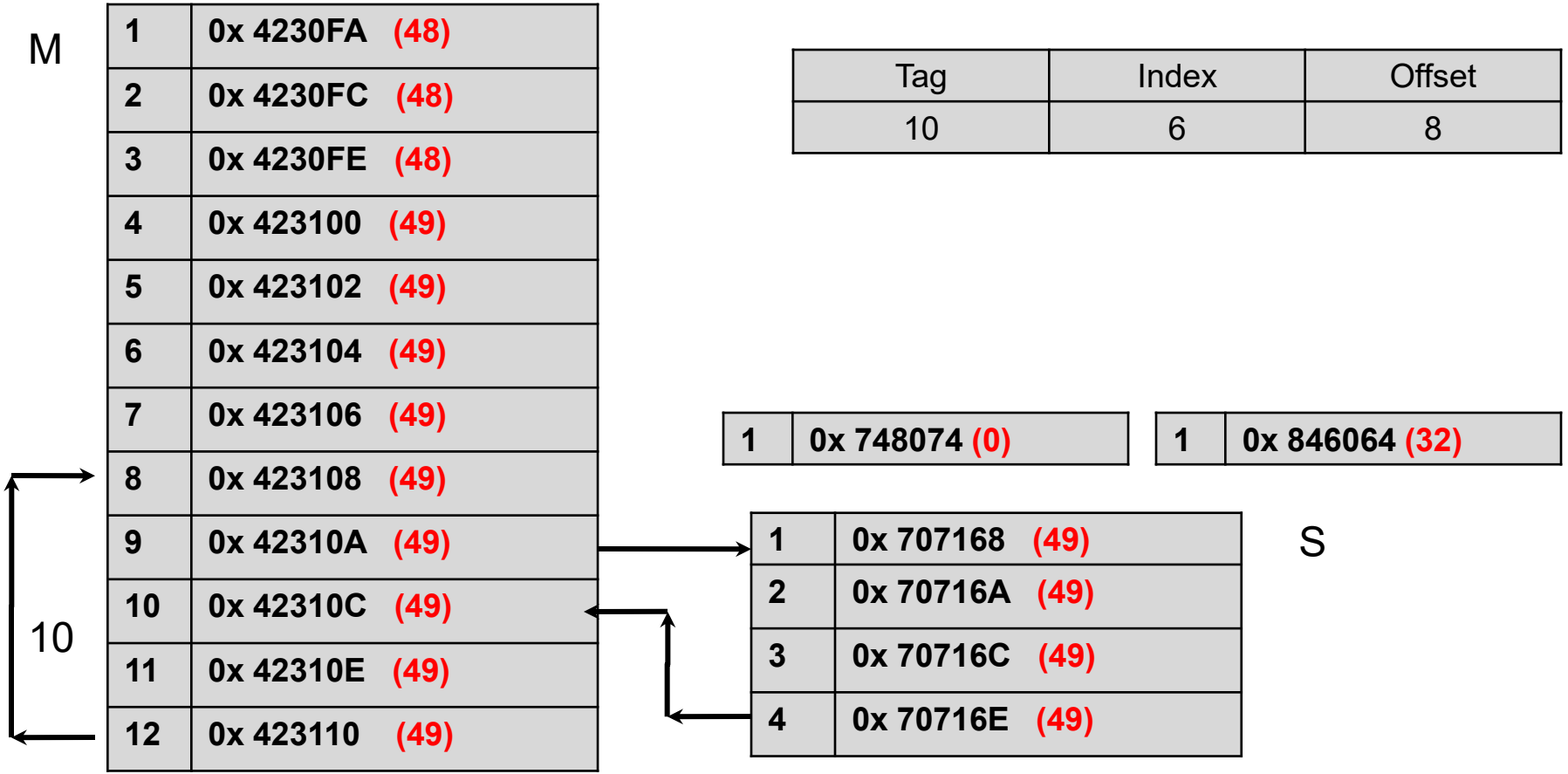
A 16KB direct mapped 256B block unified cache is attached to a 16MB main memory system. The word length as well as instruction length of the processor is 16 bits. Consider a program that consists of a main routine M which in turn calls a subroutine S. M consists of 12 instruction words which are loaded in the main memory from the address 0x4230FA onwards. The last five instructions of M is a loop that is iterated 10 times. The second instruction in the loop is a call to subroutine S. S consists of 4 instruction words loaded in the main memory from the address 0x70F168. The last instruction of S is a subroutine return back to M. The only two data words that are used by M and S are at addresses 0x748074 and 0x846064. Assume the caches are initially empty. Ignore OS level interruption and subsequent cache impact on context switching.

- Find the number of cache misses occurred during the execution of the program.
- List out the block numbers (in decimal) in the cache that are non-empty after the execution of the program

Cache Mapping by OS



Cache Mapping by OS



Cache Mapping by OS

Find the number of cache misses occurred during the execution of the program.

M1

M4

S1, M10, S1, M10, ….. (10 TIMES) = 22 MISSES for program

2 misses for data.

How many cache block evictions happened during the execution of the program?

20 EVICTIONS

List out the block numbers (in decimal) in the cache that are non-empty after the execution of the program.

Non empty blocks: 0,32,48,49



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